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Medisense: An Advanced Health Tech application Using Generative AI

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Abstract—**Many people find difficulty in understanding medical information, including complex medical reports. They don't have access to real time health information which makes it difficult for them to understand their health condition and take care of their health. Also, they don't fully understand their medications or medical conditions. To overcome these issues, a complete web-based solution, MediSense is proposed in this paper using Generative AI to simplify medical information. Medisense, a web-based solution comprises of 4 major components namely MedEase Report Simplifier, MediBot Advisor, LiveWell Monitor and RxInsight Analyzer. MediSense makes healthcare information easy to understand, answers all the questions, helps people keep track of their current health, and supports better health management. MediSense helps by providing solutions such as MedEase Report Simplifier to make medical reports easy to understand. MediBot Advisor which is an AI chatbot that answers health questions and provides information. LiveWell Monitor to track health related data from wearable devices. RxInsight Analyzer which explains prescriptions, including what each medicine does, side effects, costs, and alternatives. In our web application, we used Next.js and Tailwind CSS for the frontend, Flask and Langchain for the backend, and Gemini as the large language model. The proposed system is successfully implemented, tested and proved to be simple to use. Medisense helps people to understand healthcare information and make better health decisions.**

Keywords— Chatbot, Healthcare, Generative AI, Medibot, Report Simplifier.

I. INTRODUCTION

In today's complex healthcare landscape, patients face significant challenges in understanding medical information, accessing reliable advice, maintaining real-time health monitoring, and comprehending prescription details [1]. By making healthcare data more accessible and understandable, one can empower individuals to take control of their health, fostering a more informed and proactive approach to personal health management.

MediSense is the solution that revolves around the healthcare domain. There are few issues in the current healthcare system. MediSense aims to overcome these issues. However, when it comes to people's health, they blindly trust the advice given by the healthcare professionals, this is because they don't have access to medical information, don't understand medical terms, or don't have expert advice. Making medical information accessible to people, providing real time medical assistance 24 x 7, making people understand complex medical terminologies and developing the sense of understanding their health condition is what MediSense aims for. Understanding complex medical reports is difficult for individuals, getting real time recommendations and analysing personal health data is difficult for people, having an assistant that can provide health related data is often available but we cannot guarantee the authenticity [2][3]. Also, the healthcare system is corrupted to an extent where people are forced to buy costly medicines. MediSense is developed to address all such issues in the current healthcare system. Empowering people with medical knowledge, supporting and assisting them in making their health -related decisions is what MediSense works for.

II. RELATED WORK

The paper [4] highlighted that the patients received accurate and precise predictions of diseases based on the symptoms details provided by the user through the medibot i.e. medical chatbot. There were two arrangement calculations especially employed for forecasting sickness known as Decision Tree algorithm and K-Nearest Neighbors (KNN) algorithm. Decision Tree was used in the chatbot which gives treatment recommendation and the guidance of diagnosis by following a series of branching questions and conditions based on input given by the user. This medical chatbot provided a human element to the conversation which may feel user-friendly to the user[4][5]. An interactive chatbot enabled the retrieval of symptoms provided by the user. This approach was incorporated into existing strategies in the field of healthcare, which will provide assistance to healthcare professionals and most importantly to the patients[6].

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L. Athota el. at. [7] proposed an idea of developing a medical chatbot using Artificial Intelligence that could diagnose the diseases and which provides basic details of the disease. This chatbot was designed to improve accessibility to medical knowledge among the users. Natural Language interactions with the users were developed through computer programs used by the chatbots. The chatbot stored the data of the healthcare information in the database for the identification of the sentence keywords given by user input and answers all questions by making query decisions. By using the concept of the n-gram, TF-IDF and cosine similarity, ranking and calculation of the sentence and word similarity is performed. A score was generated for the input sentence given by the user for each sentence and from the query given similarity of sentences were generated. Questions given by the user were answered quickly[8].

C. V. S. Avila el.at.[9] explained that Medibot was a chatbot for querying drug information. This system acted as a single access point for natural and simplification of the information of drugs, it's prices and its side effects. This chatbot consists of two modes of operations: first is Quick Response and second is Interactive modes. Quick Response answered the questions which were asked in natural language and Interactive modes had three interactive tasks: Price Comparison, Query, Browser. This chatbot had a special task which presents the best prices for medicines and their key alternative substitutes identified in real-time obtained from the Web with the help of information contained in a linked data mashup. This task helps in providing best prices and alternatives for specific medicines. Best prices of medicines task achieves a success of 90% but for alternatives suggestions it gives 33% success so it indicates that improving the task of alternative suggestions. Future scope is that medical specialists from different nationalities are required to improve chatbots for implementing a translation mechanism [10].

The paper[11] explained that the design of the system consists of the integration of the real time continuous and long term health monitoring system used to monitor the biomedical parameters like body temperature, blood pressure, pulse oximetry, heart rate and ECG using biomedical sensor in the patient. The sensors were attached to the patient body, biomedical parameters were acquired from sensors. Arduino Uno controller board processes the biomedical parameters and was transmitted to a healthcare centre via Wi-Fi module. Comparison between received biomedical parameters and predefined threshold values is done. Biomedical parameters of the patients were stored in the database if their parameters were normal as of the healthy patient. Author proposed that the important role of saving the life of the human is possible only through the real time monitoring and immediate health examination of the patient. Early detection and pre-diagnosis of many health concerns was proposed in this system.

L. Bercken el.at. [12] highlighted the important point that Health Literacy was a relevant component of public health. Medical texts were difficult to understand by people of non-medical background. These texts were only understood by medical background and non-medical background people with high-skill knowledge. Dataset used in this machine translation model was created by filtering aligned health sentences using advanced knowledge. After training this dataset, it compares with models trained on a general simplification dataset using an automatic evaluation and detailed expert evaluation. An open corpus was introduced which was widely available on Wikipedia-Simple Wikipedia text simplification corpus, further this corpus is augmented with additional aligned sentences which focuses on the medical domain. Medical experts had helped in creating this corpus. This corpus was used to generate two Neural Machine Translation models: first was trained on the aligned Wikipedia corpus, and second was trained in addition on corpus but with medical texts which were replaced by the UMLS Concept Unique Identifiers. Furthermore, Training a Neural Machine Translation model on an unspecialized corpus which had an ability to generate acceptable results in a medical domain texts which hints potential of future aspirations[13].

Mirza Moiz Baig et.at. [14] highlighted the main point people were scared of visiting hospitals during COVID-19 pandemic which resulted in the lack of proper medication. Many technologies were implemented in the medical domain, one of them is Medical Chatbot. This medical chatbot was an automatized system which uses natural language diagnoses to provide medical aid. Patients could monitor his/her health using these chatbots. Development of chatbots is done using the concepts of AI, Ml, NLU, NLP etc. This paper reviewed the technologies and applications of various chatbots implemented in recent years. NLP techniques were used for processing user inputs that were formatted in strings so the program could process it. If input was not written in proper format it can be a problem for chatbots. When user inputs were tokenized, further they were processed with machine learning such as classification to process these symptoms to match the disease available in the classification training. NLP and Machine Learning were most suitable for implementing the chatbots. Upcoming years, these chatbots will have great future scope where they will directly connect to the doctor through video call for medical counseling. These chatbots will provide location for the nearest hospitals and will be available in various languages so everyone can use it.

III. PROPOSED METHODOLOGY

The proposed MediSense architecture consists of four main components, each component is designed to address specific healthcare needs: simplifying medical information, answering health-related queries, providing medication insights, and monitoring real-time health data. The figure 1 and 2 provides the architecture diagram of RAG Application and its components. These components use smart AI techniques like Retrieval Augmented Generation (RAG), semantic search, and graph database, combined with wearable devices for real-time data collection and analysis. The system offers a user-friendly interface for various medical use cases.

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A. MedEase Report Simplifier

The MedEase Report Simplifier is designed to convert complex medical reports into easy-to-understand summaries. Users can upload any medical report that is difficult to understand, and after uploading, they will receive a simplified version of the report. If user's have any query or don't understand part of the content, they can chat with the PDF directly within the system.

1) Data Collection: Medical data is collected from reliable sources such as medical books, encyclopedias, and trusted online databases.

2) Data Chunking: The collected information is divided into manageable chunks for more efficient processing and analysis. Vector Embeddings: Using Gemini's text embedding model, these data chunks are converted into vector embeddings.

3) Storage in Vector Database: The vector embeddings are stored in a vector database like ChromaDB for fast retrieval during the search process.

4) User Input Processing: User inputs (such as medical reports) are transformed into vector embeddings, which are then used to search the database.

5) Semantic Search and Information Retrieval: The system performs a semantic search by matching user input embeddings with the stored embeddings to retrieve relevant data.

6) Result Structuring: The retrieved data is organized into simplified summaries for easy understanding.

B. MediBot Advisor.

MediBot Advisor is an AI chatbot designed to answer a wide range of health-related questions integrated with medical datasets.

1) Data Collection: The system uses an extensive collection of health data from various trusted sources.

2) Vector Embeddings: Medical data is converted into vector embeddings using Gemini's model, enabling semantic search for chatbot responses.

3) Storage in Vector Database: The embeddings are stored in ChromaDB, which allows the chatbot to quickly retrieve relevant answers.

4) User Query Processing: User queries are processed into vector embeddings and matched against the stored embeddings to find the most relevant answers.

5) Contextual Responses: The chatbot retrieves the top matching results and provides responses in simple language to ensure clarity and relevance.

C. RxInsight Analyzer

RxInsight Analyzer offers detailed information about prescribed medications, including their uses, side effects, and alternatives. Users simply need to type in the brand name of the medicine, and they will receive a list of alternatives that contain the same generic ingredients.

1) Data Collection: Medicine information, including drug purposes and side effects, is gathered from reliable medical sources.

2) Graph Database (Neo4j): The medicine data is stored in a Neo4j graph database, enabling the clear representation of relationships between medications.

3) User Input Processing: User queries regarding their prescribed medications are processed and matched to the data in the graph database.

4) Graph Search and Retrieval: RxInsight performs graph based searches to find and retrieve relevant medication information, displaying the relationships between medicines, alternatives, and side effects.

5) Result Presentation: The system provides detailed information on prescribed medications, improving user understanding and aiding in decision-making.

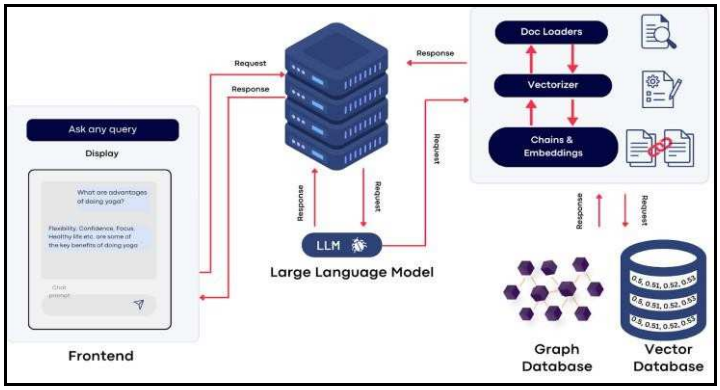


Fig. 1. Architecture Diagram of RAG Application.

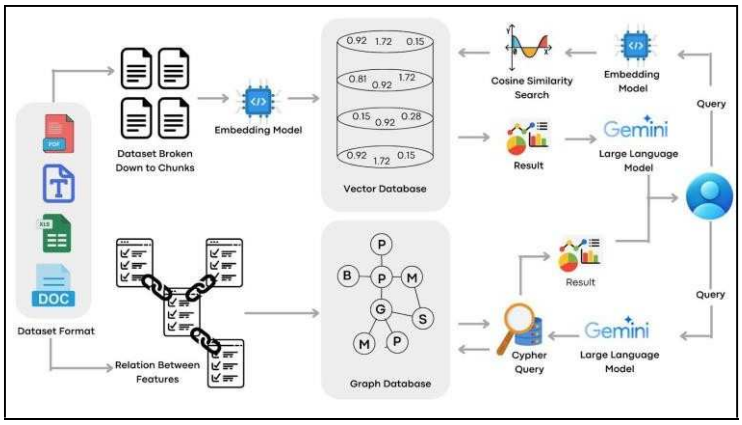


Fig. 2. Architecture Diagram of MedEase Report Simplifier, MediBot Advisor and RxInsight Analyzer

D. Live Well Monitor

LiveWell Monitor is responsible for real-time health tracking and providing personalized health recommendations based on wearable device data.

1) Data Collection: Real-time health data is collected from wearable devices like Fitbit using the Fitbit API.

2) Data Processing: The collected data is processed and analyzed to detect trends in the user's health parameters, such as heart rate, steps, and sleep patterns.

3) Personalized Recommendations: Health scores are analyzed using the Gemini API, which provides personalized health recommendations based on the data.

4) Graphical Representation: The processed data is displayed in graphical format, allowing users to monitor

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their health trends over time. The architecture diagram of LiveWell Monitor is seen in figure 3.

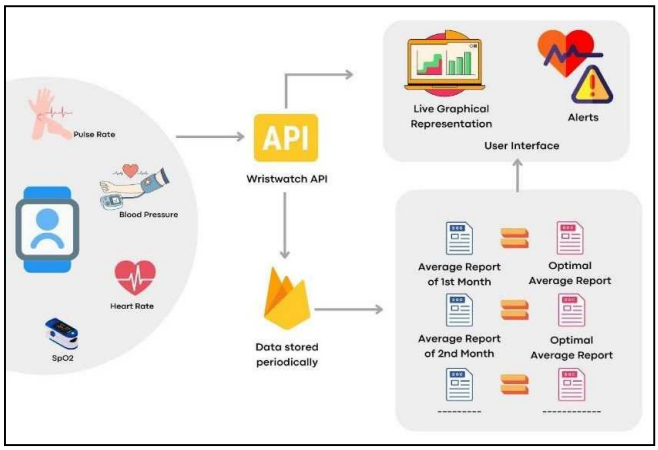


Fig. 3. Architecture Diagram of LiveWell Monitor.

IV. Result and Discussion

The MediSense is tested with various datasets and user inputs to simplify medical information, provide accurate health advice, monitor real-time health data, and analyze prescriptions. Here for implementation we used Next.js and TailwindCSS for the frontend. Flask, LangChain, Gemini, and Elevenlabs for backend logic. ChromaDB for vector storage, while Neo4j managed the graph database, and Firebase for managing authentication and user data. The Fitbit API collected health data like SpO2 and heart rate for the LiveWell Monitor. The system was deployed using Vercel for the frontend and Render for the backend and it was tested on a server with 16GB of RAM and an Intel i5 processor for optimal performance.

Figure 4 and 5 represent the MediSense homepage, which provides a detailed overview of MediSense features. It introduces the MedEase Report Simplifier for summarizing medical reports, the MediBot Advisor for answering health-related queries, the LiveWell Monitor for tracking real-time health data, and the RxInsight Analyzer to provide detailed information about prescribed medicines.



Fig 4: MediSense Homepage



Fig 5: MediSense Page

A. *MedEase Report Simplifier*

The MedEase Report Simplifier successfully convert complex medical reports into simplified and understandable language. Figure 6 demonstrates the User Interface of MediEase Report Simplifier. Users can also interact with the PDF directly through the chat section. The summary of the report is shown in figure 7.

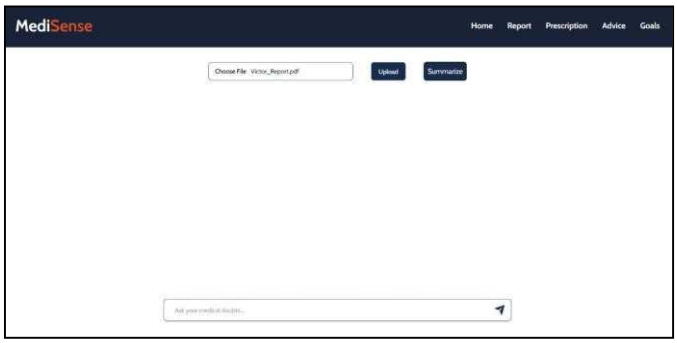


Fig. 6. User Interface of MediEase Report Simplifier

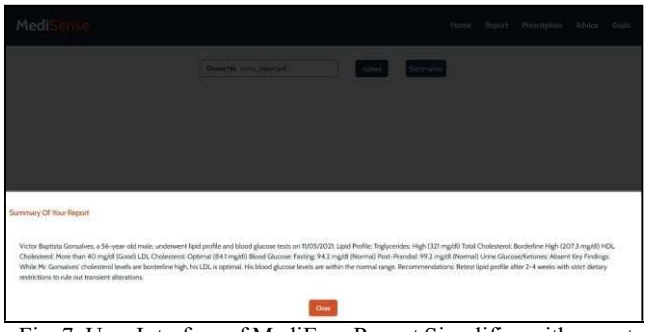


Fig. 7. User Interface of MediEase Report Simplifier with report Summary

B. *MediBot Advisor*

MediBot Advisor is a chatbot that answers health-related questions. The chatbot's integration with various datasets and the use of vector embeddings ensured accurate and relevant answers. The figure 8 displays user Interface of MediBot Advisor.



Fig. 8. User Interface of MediBot Advisor

C. *Live Well Monitor*

The LiveWell Monitor provide real-time health monitoring based on data collected from wearable devices. The system displays the health data graphically for the users to track their health trends. Additionally, users received personalized recommendations based on their health score. In fig. 9, the x-axis represents time, while the y-axis represents Users health parameter values. These parameters include SpO2 with a nor- mal range between 95 to 100, pulse rate with a normal range between 60 to 100 beats per minute, and body temperature with a normal range between 35°C to 40°C.

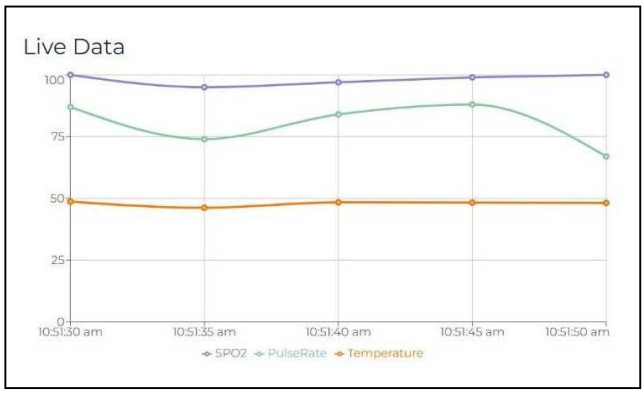


Fig. 9. User Interface of LiveWell Monitor.

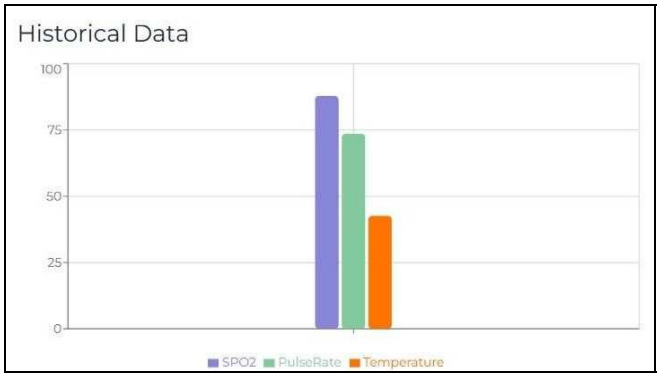


Fig. 10. User Interface of LiveWell Monitor

D. *RxInsight Analyzer*

The RxInsight Analyzer provided detailed information about prescribed medicines, including their purposes, side effects, and alternatives. The interface for the same can be seen in figure 11. The use of a graph database (Neo4j) allowed for the clear representation of relationships between different medicines, making it easier for users to choose alternative options.

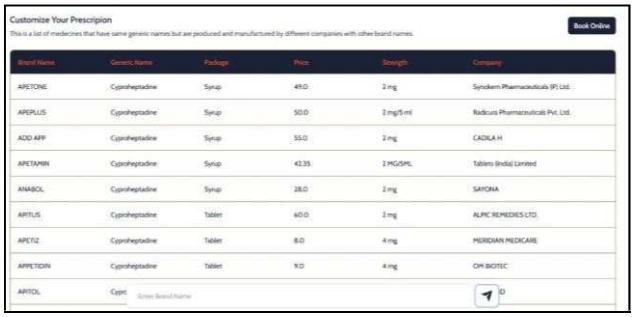


Fig. 11. User Interface of RxInsight Analyzer

V. Conclusion

The MediSense system effectively addresses the common challenges in healthcare which includes understanding complex medical information, tracking real-time health data and managing prescriptions. By implementing advanced AI techniques such as Retrieval Augmented Generation and graph database, MediSense provides high performance and accuracy. The MedEase Report Simplifier and MediBot Advisor enhance health literacy by providing users with easy access to medical information and health advice. The integration of vector embeddings and semantic search ensures that precise and relevant responses are given to the user queries. LiveWell Monitor's real-time data processing and personalized recommendations help users to manage their health. The RxInsight Analyzer further supports informed decision-making by offering detailed insights into prescribed medications, contributing to overall improved healthcare outcomes for users.

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